

Supplementary Material

Anonymous ACCV 2026 Submission

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This supplementary material complements the main paper and is organized as follows:

1. per-action quantitative comparisons on Human3.6M (Sec. A);
2. additional qualitative results on in-the-wild videos (Sec. B).

A Per-action Error Comparison

To assess the effectiveness of our method for each specific action, we present in Tab. 1 a per-action comparison of the MPJPE (P1) and P-MPJPE (P2) error rates against various alternative methods on the Human3.6M dataset, using detected 2D pose sequences. Our findings indicate that STAMPose-L achieves the best P1 error on numerous actions, including Directions, Eating, Photo, Pose, Wait, WalkDog, Walk, and WalkTogether, and attains the lowest average P1 (37.9 mm). For the remaining actions, it stays close to the best results, whereas the sitting-type actions (Sitting, SittingDown) remain the most challenging. Overall, these results show that STAMPose maintains a competitive edge across a wide range of activities.

B Additional Qualitative Results on In-the-Wild Videos

To assess generalization beyond the studio conditions of the benchmark datasets, we apply STAMPose-L to in-the-wild videos that are unseen during training. Because ground-truth 3D annotations are unavailable for such videos, we visualize only the predicted 3D poses, with the left and right body sides drawn in red and blue, respectively.

Figure 1 shows the results across a range of everyday and athletic activities, including dancing, weightlifting, and basketball. Despite motion blur and large pose variations, STAMPose recovers structurally plausible 3D poses across these diverse in-the-wild scenes.

Figure 2 focuses on harder cases where the 2D detector visibly fails, using figure-skating, taekwondo, and snowboarding clips. Red arrows mark implausible 2D detections, such as abnormal limb lengths or swapped joints under fast motion, and missing joints under occlusion in the snowboarding clip; in many such cases, STAMPose still recovers a coherent 3D pose (green arrows). The rightmost column (taekwondo) shows a failure case: under an extreme high-kick pose the 2D input is severely degraded, and this error propagates to the 3D estimate (red arrow on the 3D pose). This is consistent with the limitation in the main paper that the 2D detection stage remains a bottleneck for lifting.

Table 1: Per-action MPJPE (P1) and P-MPJPE (P2) comparison on Human3.6M using detected 2D pose sequences. T is the number of input frames. Best per column in **bold**, second-best underlined.

MPJPE(P1)	T	Dire.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD	Smoke	Wait	WalkD	Walk	WalkT	Avg
MixSTE [9]	243	37.6	40.9	37.3	39.7	42.3	49.9	40.1	39.8	51.7	55.0	42.1	39.8	41.0	27.9	27.9	40.9
P-STMO [7]	243	38.9	42.7	40.4	41.1	45.6	49.7	40.9	39.9	55.5	59.4	44.9	42.2	42.7	29.4	29.4	42.8
StridedFormer [4]	351	40.3	43.3	40.2	42.3	45.6	52.3	41.8	40.5	55.9	60.6	44.2	43.0	44.2	30.0	30.2	43.7
STCFormer [8]	243	39.6	41.6	37.4	38.8	43.1	51.1	39.1	39.7	51.4	57.4	41.8	38.5	40.7	27.1	28.6	41.0
MotionBERT [10]	243	36.6	39.3	37.8	33.5	41.4	49.9	37.0	35.5	50.4	56.5	41.4	38.2	37.3	26.2	26.9	39.2
HDFormer [1]	96	38.1	43.1	39.3	39.4	44.3	49.1	41.3	40.8	53.1	62.1	43.3	41.8	43.1	31.0	29.7	42.6
HSTFormer [6]	81	39.5	42.0	39.9	40.8	44.4	50.9	40.9	41.3	54.7	58.8	43.6	40.7	43.4	30.1	30.4	42.7
MotionAGFormer-L [5]	243	36.8	<u>38.5</u>	<u>35.9</u>	<u>33.0</u>	41.1	48.6	38.0	34.8	<u>49.0</u>	51.4	<u>40.3</u>	<u>37.4</u>	36.3	27.2	27.2	38.4
PoseMamba-S [3]	243	39.5	41.6	39.9	35.7	43.5	52.7	40.2	37.9	53.1	63.4	42.4	40.0	39.6	29.3	29.2	41.8
PoseMamba-B [3]	243	38.8	40.8	38.8	35.2	42.1	50.8	38.8	36.4	51.8	61.9	42.0	38.4	38.7	28.1	28.7	40.8
PoseMamba-L [3]	243	<u>36.5</u>	37.9	37.5	33.5	39.6	47.7	<u>36.5</u>	34.0	47.8	54.5	40.2	36.3	36.7	26.4	26.9	<u>38.1</u>
SasMamba [2]	243	39.0	43.0	39.2	36.1	43.2	51.3	40.0	37.2	52.7	61.7	43.1	38.9	39.8	28.3	28.6	41.5
SasMamba-large [2]	243	38.4	41.0	38.0	34.6	41.3	49.3	38.8	36.1	50.7	56.2	41.6	38.4	37.2	27.2	27.6	39.8
STAMPose-S	243	39.6	41.7	38.0	35.1	43.0	49.7	38.6	37.5	51.2	60.0	43.0	40.3	38.2	27.4	28.8	40.8
STAMPose-B	243	37.4	39.5	38.2	32.9	41.5	<u>47.1</u>	<u>36.5</u>	35.5	49.5	<u>53.5</u>	41.7	37.6	<u>36.2</u>	<u>26.0</u>	<u>26.5</u>	38.7
STAMPose-L	243	36.0	39.1	35.7	<u>33.0</u>	<u>40.1</u>	45.3	36.0	<u>34.5</u>	49.8	55.4	<u>40.3</u>	36.3	35.3	25.5	26.1	37.9
P-MPJPE(P2)	T	Dire.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD	Smoke	Wait	WalkD	Walk	WalkT	Avg
MixSTE [9]	243	30.8	33.1	30.3	31.8	<u>33.1</u>	39.1	31.1	30.5	42.5	44.5	<u>34.0</u>	30.8	32.7	22.1	22.9	32.6
P-STMO [7]	243	31.3	35.2	32.9	33.9	35.4	39.3	32.5	31.5	44.6	48.2	36.3	32.9	34.4	23.8	23.9	34.4
StridedFormer [4]	351	32.7	35.5	32.5	35.4	35.9	41.6	33.0	31.9	45.1	50.1	36.3	33.5	35.1	23.9	25.0	35.2
STCFormer [8]	243	29.5	33.2	<u>30.6</u>	31.0	33.0	38.0	30.4	<u>29.4</u>	41.8	<u>45.2</u>	33.6	29.5	31.6	21.3	22.6	32.0
MotionBERT [10]	243	30.8	32.8	32.4	28.7	34.3	38.9	<u>30.1</u>	30.0	42.5	49.7	36.0	30.8	31.7	22.0	23.0	32.9
HDFormer [1]	96	<u>29.6</u>	33.8	31.7	31.3	33.7	<u>37.7</u>	30.6	31.0	<u>41.4</u>	47.6	35.0	30.9	33.7	25.3	23.6	33.1
HSTFormer [6]	81	31.1	33.7	33.0	33.2	33.6	38.8	31.9	31.5	<u>43.7</u>	46.3	35.7	31.5	33.1	24.2	24.5	33.7
MotionAGFormer-L [5]	243	31.0	<u>32.6</u>	31.0	27.9	34.0	38.7	31.5	30.0	<u>41.4</u>	45.4	34.8	30.8	31.3	22.8	23.2	32.5
PoseMamba-S [3]	243	32.1	34.8	34.1	30.7	36.0	41.7	33.0	32.1	44.5	53.8	36.5	32.3	34.2	25.0	25.2	35.0
PoseMamba-B [3]	243	32.3	34.0	33.2	30.2	34.9	40.6	32.0	31.0	44.5	53.0	36.3	31.3	33.5	23.8	24.6	34.3
PoseMamba-L [3]	243	31.0	32.2	32.5	28.5	<u>33.1</u>	38.6	30.5	29.2	41.3	47.8	34.9	30.2	31.7	22.7	23.5	32.5
SasMamba [2]	243	32.3	35.7	33.5	30.8	36.4	40.2	32.8	32.1	44.0	52.7	37.0	31.7	34.3	24.4	24.8	34.8
SasMamba-large [2]	243	31.9	34.0	32.4	29.8	34.7	39.2	31.7	31.3	43.2	49.5	36.2	31.1	32.2	23.5	23.9	33.6
STAMPose-S	243	33.0	34.5	32.8	30.0	35.8	38.9	31.7	32.2	43.4	52.8	37.1	32.2	32.7	23.2	24.5	34.3
STAMPose-B	243	31.7	33.4	32.8	<u>28.4</u>	35.0	38.0	<u>30.1</u>	31.3	42.9	47.5	36.5	30.7	<u>31.1</u>	22.3	23.2	33.0
STAMPose-L	243	30.8	32.9	31.0	<u>28.4</u>	33.0	37.2	29.9	30.1	42.6	47.5	35.0	<u>29.6</u>	30.3	<u>21.7</u>	<u>22.7</u>	<u>32.2</u>

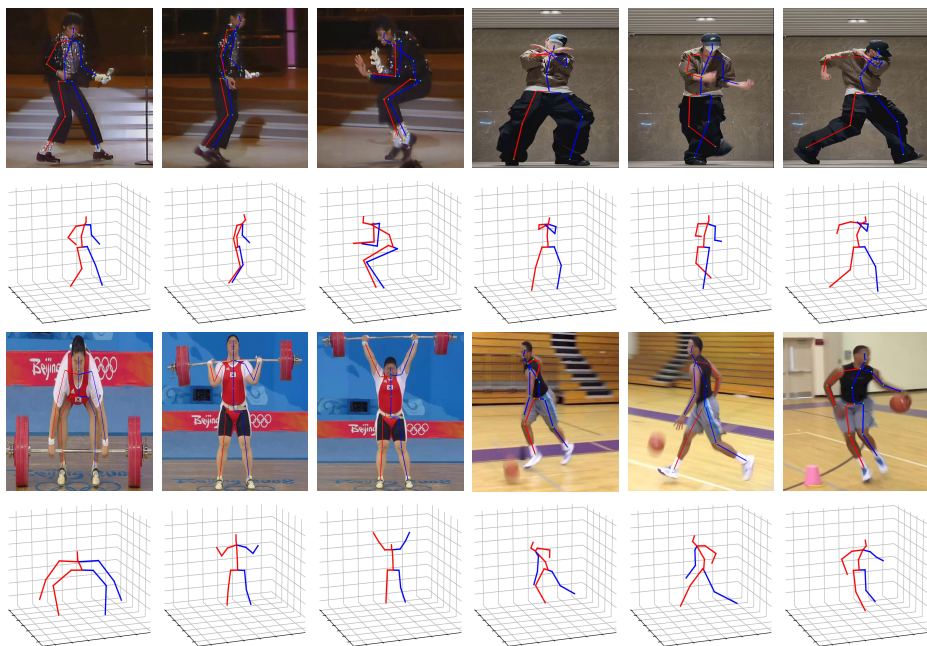


Fig. 1: Qualitative results of STAMPose on in-the-wild videos.

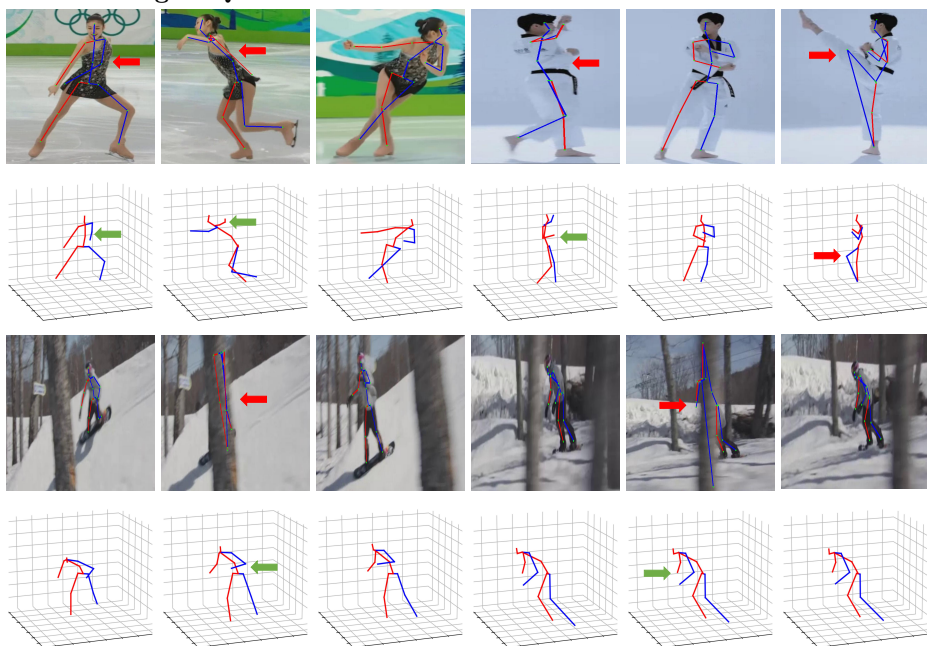


Fig. 2: Qualitative results of STAMPose on in-the-wild videos where the 2D detector fails.

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